CASE STUDY

Exploration of Cloud Service Models (IaaS, PaaS, SaaS) Using a Free-Tier Cloud Platform

1. Introduction

Cloud computing allows users to access computing resources such as servers, storage, databases, networking, and software through the Internet. Instead of purchasing and maintaining physical infrastructure, users can use cloud services according to their requirements.

Cloud computing is commonly classified into three major service models: Infrastructure as a Service (IaaS), Platform as a Service (PaaS), and Software as a Service (SaaS). IaaS provides virtual machines, storage, and networking resources. PaaS provides an environment for developing, testing, and deploying applications. SaaS provides ready-to-use software through a web browser or application.

Free-tier cloud platforms allow students to explore cloud computing concepts with limited resources or free usage allowances. This case study uses examples from AWS and browser-based software to understand the features and differences between the three service models.

2. Aim

To explore and compare IaaS, PaaS, and SaaS using a free-tier cloud platform and understand their applications, advantages, and limitations.

3. Objectives

- Understand the basic concepts of cloud computing.
- Study the characteristics of IaaS, PaaS, and SaaS.
- Explore cloud services available through a free-tier platform.
- Understand the differences between infrastructure, application development platforms, and ready-to-use software.
- Compare the advantages and limitations of the three service models.
- Identify suitable cloud service models for different applications.

4. Problem Statement

Setting up and maintaining physical servers, software environments, and storage systems can be expensive and time-consuming. Students and developers may not have access to the resources required to experiment with different computing environments. Cloud computing provides an alternative by offering infrastructure, development platforms, and software over the Internet.

5. Selected Cloud Platform

Example platform: Amazon Web Services (AWS)

AWS provides services for computing, storage, databases, application deployment, and networking. Users can explore eligible services through available free offers, subject to account eligibility, usage limits, and current pricing conditions. Microsoft Azure and Google Cloud also provide cloud services for learning and experimentation.

6. Cloud Service Models

6.1 Infrastructure as a Service (IaaS)

IaaS provides virtual computing infrastructure over the Internet. Users can create virtual machines, configure operating systems, manage storage, and configure networking.

Example: Amazon EC2 virtual machines.

Applications: Hosting websites and web servers; running operating systems and applications; creating testing environments; storing and processing data.

Advantages: Flexible and scalable resources, greater operating-system control, and reduced need to purchase physical servers.

Limitations: Users must manage operating systems and updates; security configuration requires care; usage beyond eligible free allowances may incur charges.

6.2 Platform as a Service (PaaS)

PaaS provides a platform for developing, testing, and deploying applications without requiring users to manage the underlying infrastructure in detail.

Example: AWS Elastic Beanstalk.

Applications: Web application development, application testing and deployment, rapid software development, and managed application environments.

Advantages: Simplifies deployment, reduces infrastructure management, and supports faster development.

Limitations: Less control over infrastructure; platform restrictions may affect design; associated resources may incur charges.

6.3 Software as a Service (SaaS)

SaaS provides ready-to-use software through the Internet. The service provider manages the application and its underlying infrastructure.

Examples: Google Docs, Microsoft 365 online applications, and Salesforce.

Applications: Online document creation, email and communication, customer relationship management, and file sharing.

Advantages: Browser access, minimal installation and maintenance, and convenient collaboration.

Limitations: Internet access is often needed; customization may be limited; features and storage may depend on subscription or usage policies.

7. Basic Architecture

- **IaaS:** The user manages the operating system and applications; the provider manages physical infrastructure.
- **PaaS:** The user manages application code and data; the provider manages the platform and underlying infrastructure.
- **SaaS:** The provider manages the application and infrastructure; the user accesses the software.

The exact division of responsibilities depends on the selected service and its configuration.

8. Requirements

- Computer or laptop.
- Internet connection and web browser.
- Cloud platform account, if required.
- Access to eligible free-tier services or free software alternatives.

9. Procedure

Step 1: Access the cloud platform. Open the official AWS website and create or sign in to an account if eligible. Review current free-tier terms, available services, and billing conditions.

Step 2: Explore IaaS. Open Amazon EC2 in the AWS Management Console. Examine virtual machine options, operating system images, storage, and networking settings. Study the configuration before launching a resource. If you launch one, choose an eligible option and verify estimated charges and cleanup steps.

Step 3: Explore PaaS. Open AWS Elastic Beanstalk. Study how an application can be uploaded and deployed through a managed application environment. Observe the relationship between application code, the deployment platform, and supporting infrastructure.

Step 4: Explore SaaS. Open a browser-based application such as Google Docs. Create a sample document, edit its contents, and observe online access and collaboration.

Step 5: Compare the models. Compare infrastructure management, application development, user control, maintenance, and common use cases.

Note: This is an exploration activity. Service availability and free usage eligibility may vary by account and region.

10. Comparison of Cloud Service Models

Criteria	IaaS	PaaS	SaaS
Full form	Infrastructure as a Service	Platform as a Service	Software as a Service
Main purpose	Computing infrastructure	Development and deployment environment	Ready-to-use software
Example	Amazon EC2	AWS Elastic Beanstalk	Google Docs
User control	High	Medium	Limited
Management	User manages OS and apps	User manages code and data	Provider manages app and infrastructure
Typical users	Administrators and developers	Application developers	General users and organizations
Main benefit	Flexibility and control	Simplified development	Ease of use

11. Observations

- IaaS provides greater control over virtual computing resources.
- PaaS simplifies application deployment.
- SaaS provides ready-to-use software without infrastructure management.
- The appropriate model depends on technical requirements and desired control.

12. Results

The exploration demonstrated the differences between IaaS, PaaS, and SaaS.

- IaaS was studied as a model providing virtual machines, storage, and networking.
- PaaS was studied as a model supporting application development and deployment.
- SaaS was studied as a model providing ready-to-use software over the Internet.
- The advantages, limitations, and responsibilities of each model were identified.
- The comparison showed that the best service model depends on user requirements and the level of control needed.

The activity provides a basic understanding of how cloud services support computing, software development, and online collaboration.

13. Advantages of Cloud Computing

1. Reduces the need to purchase physical computing equipment.
2. Provides access to computing resources over the Internet.
3. Supports scalability according to application requirements.
4. Enables collaboration and remote access.
5. Offers different levels of control through IaaS, PaaS, and SaaS.

14. Limitations

1. Internet connectivity may be required to access cloud services.
2. Free-tier services have usage limits and eligibility conditions.
3. Exceeding free allowances may result in charges.
4. Data privacy and account security must be managed carefully.
5. Availability and customization depend on the selected provider.

15. Conclusion

This case study explored the three major cloud computing service models: IaaS, PaaS, and SaaS. IaaS provides virtual infrastructure, PaaS simplifies application development and deployment, and SaaS provides ready-to-use applications through the Internet.

The comparison highlighted differences in user control, infrastructure management, application development, and maintenance responsibilities. Free-tier cloud offerings can provide students with a useful starting point for learning cloud computing, provided they understand the applicable limits and billing conditions.

Thus, the experiment establishes a basic understanding of cloud service models and their practical applications in modern computing.

Practical evidence (if required): Add screenshots of the EC2 service page, Elastic Beanstalk page, and Google Docs. Only report a resource as launched or deployed if that step was actually completed.